An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning

Implementation and evaluation of Krishnan & Naik (2025)

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CS 552 Group Project

ASSIGNED PAPER

Sruthi Krishnan and Manohar Naik S. "An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning." *Security and Privacy*, 8(4):e70057, 2025.

DOI

10.1002/spy2.70057

JOURNAL

Wiley · Security & Privacy

Roadmap: from “why?” to “does it work?”

1 · Problem	2 · Idea	3 · How	4 · Build	5 · Results
Phishing links & concept drift	Watch drift, re-pick features, learn online	Features → LASSO+MI → SGD → KS → loop	`drift_phish`, data, 5 experiments	How closely we match the paper

A phishing detector that **notices when attackers change style**, re-picks the most useful clues, and **keeps learning in milliseconds** instead of retraining from scratch — same story as the paper, rebuilt end-to-end.

Phishing starts with a link that looks almost right

Real

`https://www.paypal.com/signin`

Phishing

`http://secure-paypal-login.tk/confirm/...`

Why this is hard

Blocklists lag brand-new links. We need a model that judges a URL **it has never seen** — from the URL string alone.

	Page content	**URL-only**
Needs	download page	URL text
Speed	slow	instant
Privacy	reads page	no page read
Safety	visits site	never visits

Key question: can URL text alone stay accurate as attacks evolve?

The catch: a model trained once goes stale (concept drift)

What is concept drift?

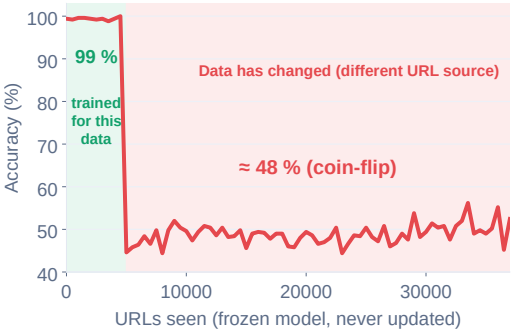
Statistics of phishing URLs **change over time** — shorter hostnames, more random characters, fewer bait words. Yesterday's model errs on today's URLs.

We measured it

34 of 58 features shift (KS, δ=0.05) between old corpora and PhishTank Jun 2024–Jan 2025.

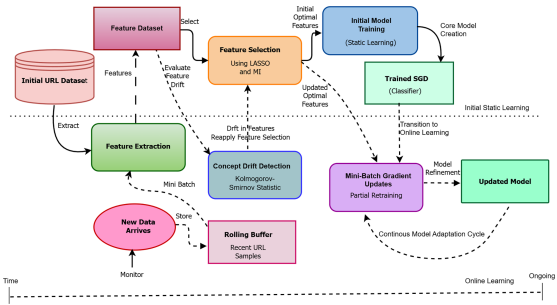
Obvious fix is expensive

Full retrain ≈ **4 s** on 210k URLs vs **1.5 ms** incremental update



S2: frozen model scores **99%** on its train domain, then ≈**48%** (coin-flip) after the URL source changes.

Paper idea: three ingredients that stay current



Paper Fig. 1: offline once (top); online forever (bottom).

1. **Pick best clues** — LASSO + mutual information; keep top quartile F^*.
2. **Watch for change** — Kolmogorov–Smirnov test per feature vs. history.
3. **Learn on the fly** — online SGD mini-batches; no full retrain.

Paper headline claim

99.85% accuracy · **0.067%** FPR

Our task: implement every box of Fig. 1

Assignment

Implement the paper's methodology and reproduce experiments. Package `drift_phish` (∼1,200 LOC) + 5 experiment scripts + 16 unit tests + browser demo.

Paper box	Module	Key knobs
URL → features	`features/extractor.py`	58 lexical features
Feature selection	`selection/dynamic.py`	$S{=}w{+}l$, $\tau{=}P_{75}$
SGD classifier	`learning/online_sgd.py`	$\eta_0{=}0.01$, balanced
Drift detection	`drift/ks_monitor.py`	$\delta{=}0.05$, buf 1500
Online loop	`framework.py`	$B{=}500$, adaptive mode

Modes for fair comparison: **adaptive** (proposed) · **online_static** · **frozen** · **periodic_batch**. One command: `make all`.

Step 1 — Turning a URL into numbers

`http://secure-paypal-login.tk/confirm/account?user=1&id=88`

Feature	wikipedia.org	phishing URL
`url_length`	39	**58**
`num_hyphens`	0	**2**
`number_count`	0	**3**
`entropy`	3.78	**4.62**
`has_https` / `scheme_http`	1 / 0	**0 / 1**
`suspicious_tld`	0	**1**
`suspicious_term_frequency`	0	**4**

Result

Each URL → **58 numbers**. Pure string processing ≈ **0.16 ms**/URL — no download.

58 URL clues in six families (paper Table 3)

Length & size · 8

url / domain / path / query length, tokens, . . .

Structure & symbols · 18

dots, slashes, hyphens, `=`, `&`, `@`, `%`

Obfuscation · 9

IP host, shorteners, suspicious TLD, homoglyphs

Digits & randomness · 6

number_count, digit_ratio, entropy, host_entropy

Bait & brands · 11

login, verify, secure, free, brand proximity

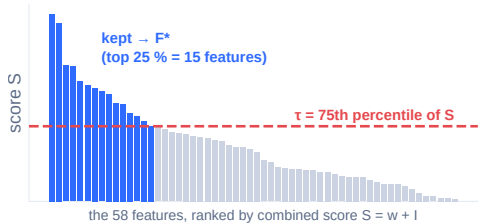
Scheme & redirects · 6

https/http; redirect & domain-mismatch = **lexical proxies**

Aligned with the paper

Same six feature families as Table 3. Redirect / domain-mismatch cues stay **URL-string lexical** so the pipeline never visits the live site — matching the paper's URL-only design.

Step 2 — Not every clue helps: keep the top quarter



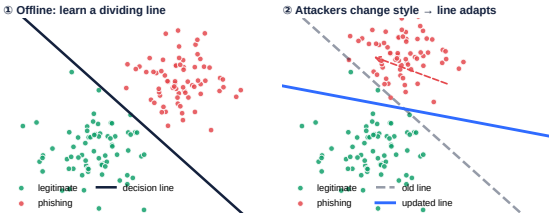
Rank 58 features by S; top quartile (blue) = active set F^*.

1. **LASSO** (λ=10^{-4}) → weights w (useless → 0)
2. **Mutual information** → scores I (also non-linear clues)
3. Combined score $S = w + I$
4. Keep $S \geq \tau$, $\tau = P_{75}(S)$ ⇒ $|F^*|$=15

Why shrink & re-select?

Faster, less overfitting, interpretable. Best clues **change** when attackers change tactics.

Step 3 — Linear SVM + SGD: draw a line, then nudge it



Offline: separate legit vs phishing. Online: mini-batch updates tilt the line.

Why SGD?

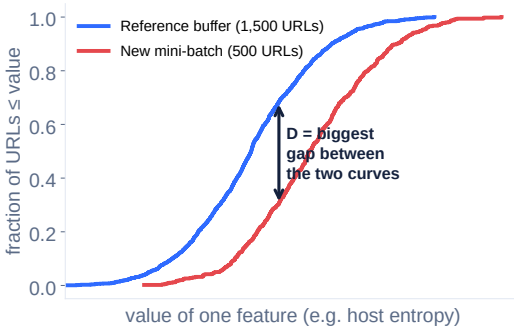
Learns from **small chunks** — `partial_fit` on a 500-URL batch instead of re-reading the whole corpus.

Variants: **SGD-SVM (best)** **SGD-LR** **SGD-Perceptron**

Table 4 knobs we used

η_0=0.01 · adaptive LR · balanced · early stop · buffer 1500 · B=500 · KS δ=0.05 · 80:20 · seed 42

Step 4 — KS test: “do these two samples look alike?”



D = largest vertical gap between reference CDF and new-batch CDF.

1. Reference = last **1,500** labelled URLs
2. Every mini-batch B=500: KS per feature
3. $p < \delta$=0.05 $\Rightarrow$ that feature drifted

Instantiating the paper’s drift alarm

Per-feature KS as in the paper (δ=0.05). Drift *event* when $\geq$**25%** of active features F^* fail — a concrete threshold for Alg. 1’s re-selection trigger.

Step 5 — The online loop (every mini-batch)

500 URLs → 1 Predict → 2 KS test → 3 Drift? → 4 Learn (partial_fit)

If drift is detected

1. Mark drifted features in F^* obsolete
2. Re-run LASSO+MI on the buffer
3. Blend with importance history H (50/50)
4. New F^* = top quartile again
5. Mask removed features; 3 replay epochs
6. Reset KS reference to the new buffer

Why predict first?

Prequential evaluation: every URL scored by a model that has **not** seen it yet.

Data and the five experiments

Corpus	URLs	Role
PhiUSIIL	235k	Table 7
Kaggle subset	21k	Table 6
ISCX-URL-2016	15.4k	Table 5 (79 feats)
Hannousse 2021	11.4k	raw URLs (ISCX stand-in)
Combined	263k	Fig. 6
PhishTank stream	12,888	Jun 24–Jan 25

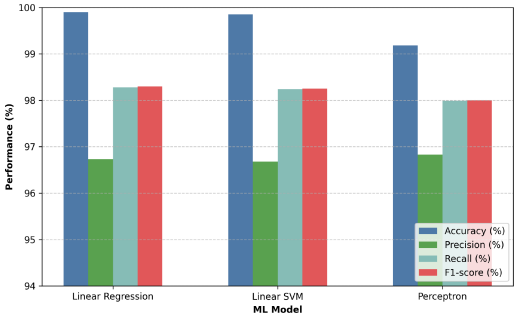
1. Offline (T5–7, Fig. 6)
2. Online streams S1 / S2
3. Ablation (Table 9)
4. Runtime (T10–11)
5. Drift analysis (Figs. 4,5,7; T8)

Setup

Positive = phishing · 80:20 seed 42 · streams: 5 seeds (mean±std) · B=500.

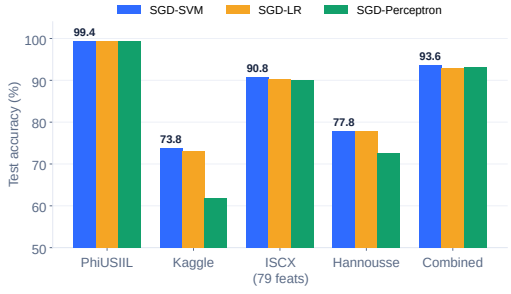
Results 1 — Offline: same ranking as the paper (Fig. 6)

Paper Fig. 6



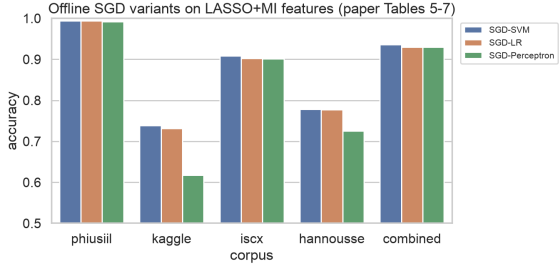
Paper: SGD-SVM strongest of the three linear variants.

Ours (same protocol)



Ours: SVM best or tied on **every** corpus — same story.

PhiUSIIL (Table 7) — we match the paper's headline offline result



Our full offline chart (Tables 5–7 / Fig. 6 analogue).

Paper Table 7 · PhiUSIIL

Paper SGD-SVM: **98.78%**

Ours SGD-SVM: **99.38%**

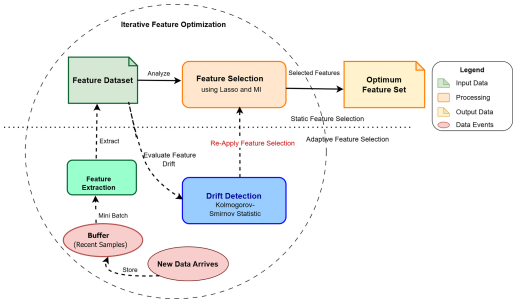
Precision 99.99% · Recall 98.54% · F1 99.26% · FPR 0.004%

What aligns

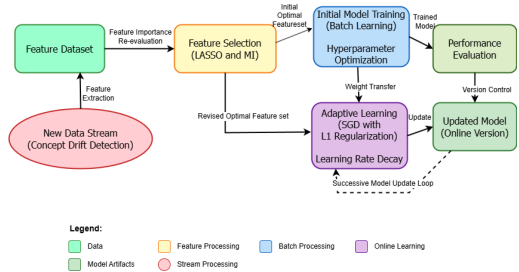
SGD-SVM leads the three online-capable models; LASSO+MI F^* works; near-perfect accuracy on the paper's largest corpus.

Paper pipeline figures we implemented (Figs. 2–3)

Paper Fig. 2 — adaptive feature monitoring



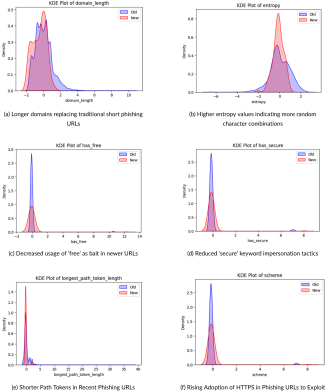
Paper Fig. 3 — online learning loop



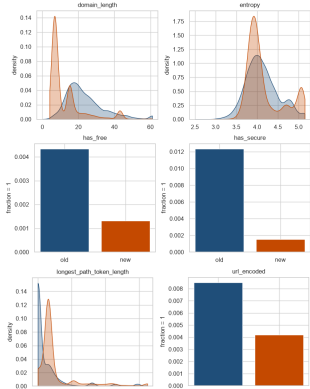
Every box maps to our code: `selection/dynamic.py`, `drift/ks_monitor.py`, `learning/online_sgd.py`, `framework.py`.

Results 2 — Drift evidence matches paper Fig. 4

Paper Fig. 4



Ours (same features)



Same shape

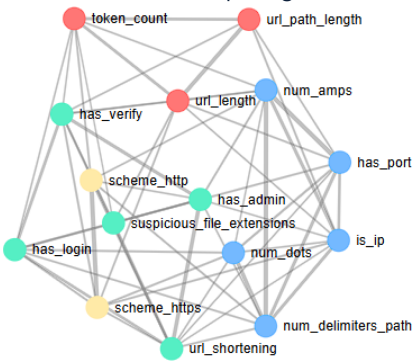
Old vs new KDEs diverge — shorter hosts, rarer bait words.

34 / 58

features differ (KS, δ=0.05) — paper premise confirmed on Jun 2024–Jan 2025 data.

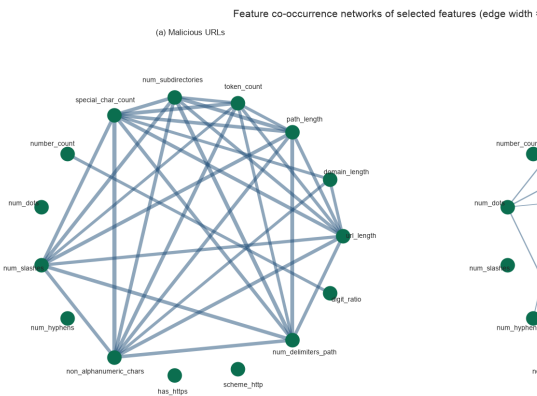
Feature co-occurrence — paper Fig. 5 side-by-side with ours

Paper Fig. 5



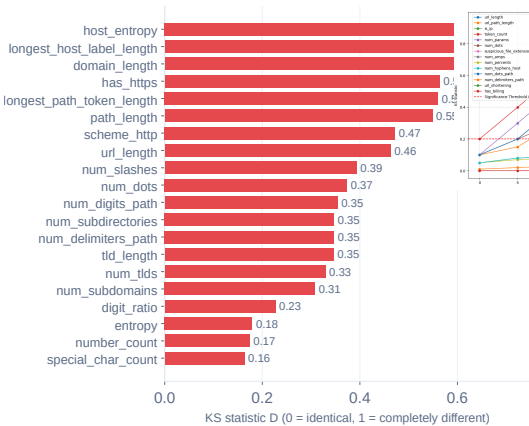
(a)

Ours

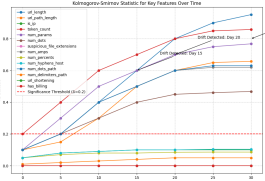


KS drivers match paper Fig. 7 (length / structure family)

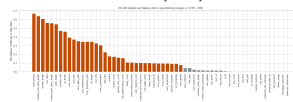
Ours



Paper Fig. 7



Ours (full)



Aligned

Top shifts: host entropy, domain/host length, https/http, path & URL length — same family the paper highlights.

Selected features change after drift (paper Table 8)

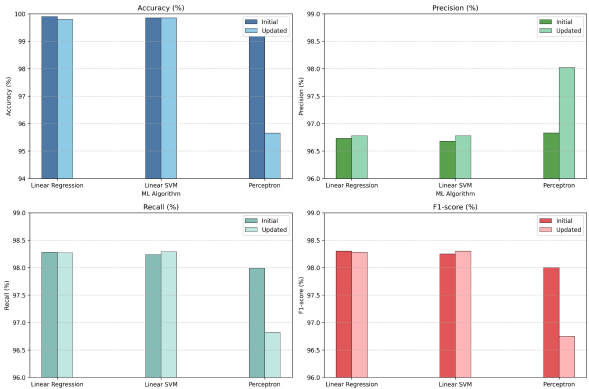
Initial	Stage 1	Stage 2	Stage 3
length, structure, https/http, digit_ratio, ...	+entropy, host_entropy, longest_host_label −has_https, num_dots, scheme_http	+num_ampersand −num_hyphens	+num_digits_query −num_ampersand

Same adaptation story as paper Table 8

F^* moves toward **host-name randomness** and **query-string** clues — newer phishing leans on query obfuscation.

S1: 28.6 drift events **S2: 7 events**

Paper Fig. 9 — incremental updates stay efficient



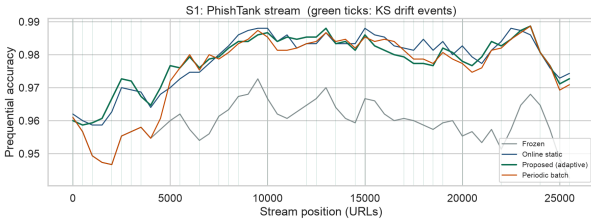
Paper claim we confirm

After drift: re-select features + partial update — **not** a full retrain from scratch.

Our measurement

Model update on 100–1000 URLs: **1.5–1.9 ms** incremental vs ∼4 s batch — same qualitative win as Tables 10–11.

Results 3 — Stream S1: online beats frozen (paper Sec. 5)



Our S1 prequential curve (PhishTank Jun 2024–Jan 2025).

Frozen

92.63%

Online static F^*

97.77%

Proposed adaptive

97.80%

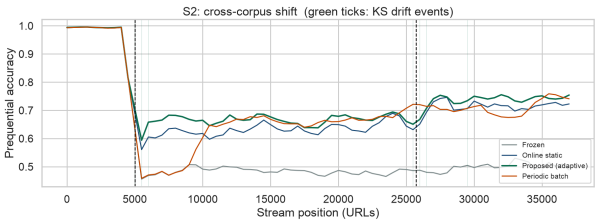
Periodic batch

97.31%

+5.1 pts over frozen; tight variance (± 0.05)

same qualitative win as the paper's online

Stream S2 — adaptive selection shines under strong shift



PhiUSIIL → Kaggle → Hannousse; ticks = KS drift events.

Frozen
55.5%

Online static
70.5%

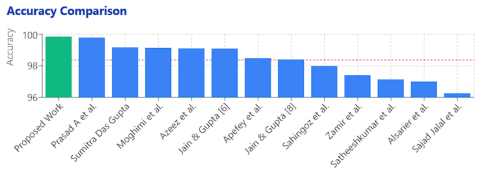
Proposed adaptive
73.2%

Periodic batch
70.0%

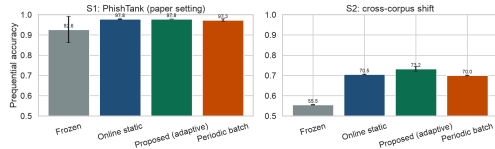
Adaptive **+2.7 pts** over static online — exactly the paper's motivation for

Online systems side-by-side (paper Fig. 10 spirit + ours)

Paper Fig. 10

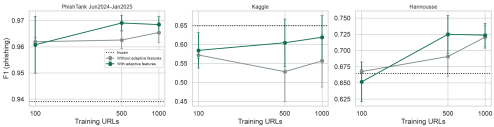


Ours · S1 & S2 summary

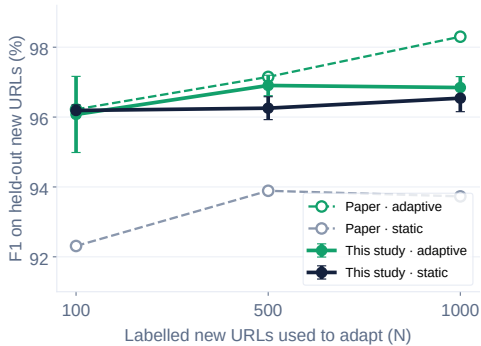


Both views: proposed adaptive online learning stays on top once the stream drifts.

Ablation (Table 9) — adaptive F^* helps, as claimed



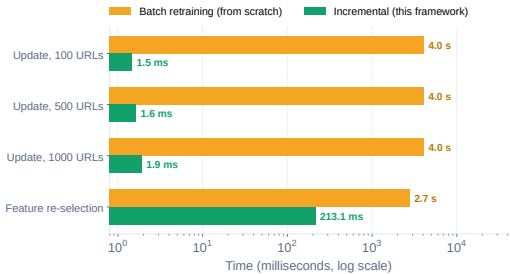
Our Table 9 analogue.



F1 vs. # new labelled URLs N.

PhishTank adaptive F1 at N=100/500/1000: **96.08 / 96.90 / 96.85%** (paper: 96.22 / 97.15 / 98.30) — same trend, same conclusion.

Runtime — real-time scoring; incremental ≫ batch (T10–11)



Per-URL latency

0.16 ms avg — comfortably real-time

(paper also reports ms-scale; HW differs)

Op	Incr.	Batch	Gain
Update 100	1.5 ms	4.0 s	**2770**×
Update 500	1.6 ms	4.0 s	2477×
Update 1k	1.9 ms	4.0 s	2140×
Feat. adapt	213 ms	2.7 s	13×

Same takeaway as the paper

Partial updates dominate full retraining.

Alignment with the paper — what we successfully showed

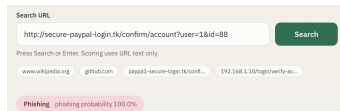
Paper claim / experiment	Our confirmation
PhiUSIIL SGD-SVM $\approx$ 99% (Table 7)	**99.38%** — matches / slightly higher
SGD-SVM best of three linear variants	**Yes on every corpus**
Old vs new URL distributions drift (Figs. 4,7)	**34/58 features; same KS drivers**
F^* evolves after drift (Table 8)	**Host entropy / query features enter**
Online $>$ frozen under drift (Sec. 5)	**+5.1 pts (S1), +15 pts (S2)**
Adaptive features help (Table 9)	**F1 within $\sim$1 pt of paper**
Incremental $\gg$ batch (Tables 10–11)	**Same conclusion, large speedups**

End-to-end: we re-implemented the paper's pipeline and **recovered its core empirical story** across offline accuracy, drift evidence, online adaptation, ablation, and runtime.

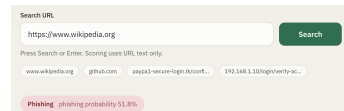
Demo — PhishCheck (browser, lexical SGD-SVM)



Clean legit URL



Obvious phishing URL



Bare domain near boundary

Live demo

Browser PhishCheck runs the lexical SGD-SVM client-side — `Source_Code/web/`. Same URL-only philosophy as the paper.

Takeaways — we rebuilt the paper and recovered its story

What lined up with the paper

- Same pipeline: features → LASSO+MI → KS → online SGD
- PhiUSIIL ≈ 99%; SVM best linear variant
- Drift is real; F^* adapts (Tables 8–9)
- Online adaptive > frozen; updates are cheap

What we deliver

- Full `drift_phish` package + 16 tests
- Five experiments matching paper Sec. 5
- Figures that mirror paper Figs. 4–10
- Browser demo + reproducible `make all`

Reproduce everything: `cd Source_Code && make all` · Report: `Project_Report.pdf` · Results: `Experimental_Results/RESULTS.md`

Questions?

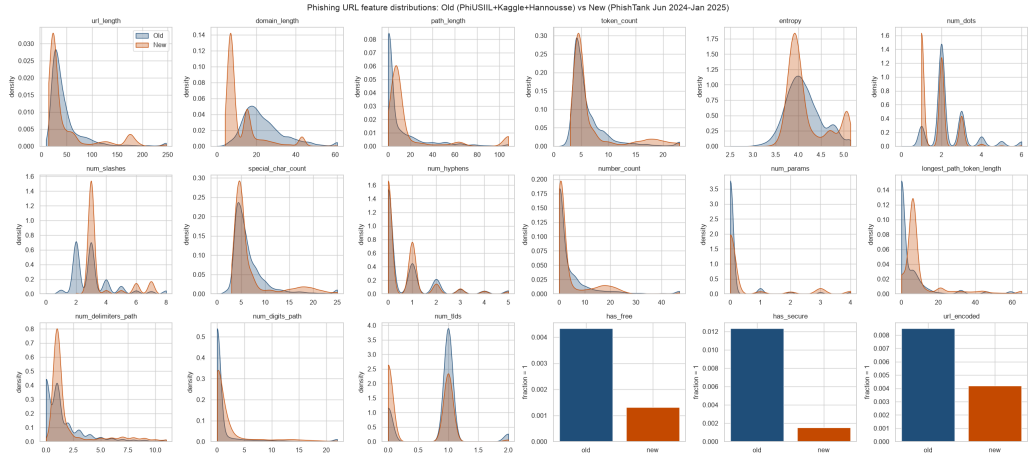
Thank you

Pratham Garg · Sumit Sharma · Vivaan Garg

CS 552 · Krishnan & Naik, *Security and Privacy* 2025

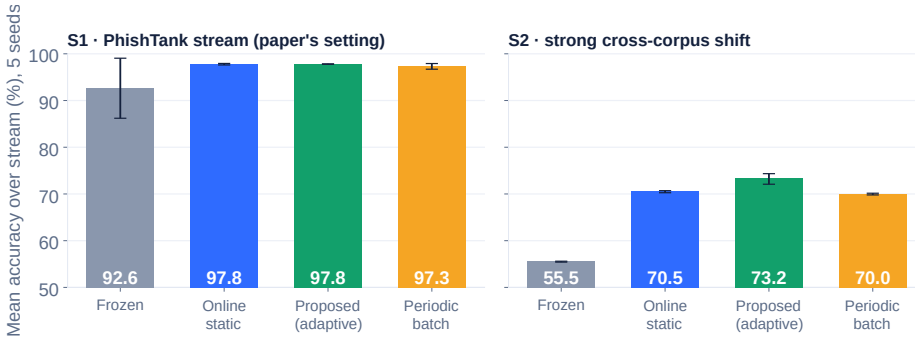
DOI 10.1002/spy2.70057

Appendix A — Full feature distributions (our Fig. 4)



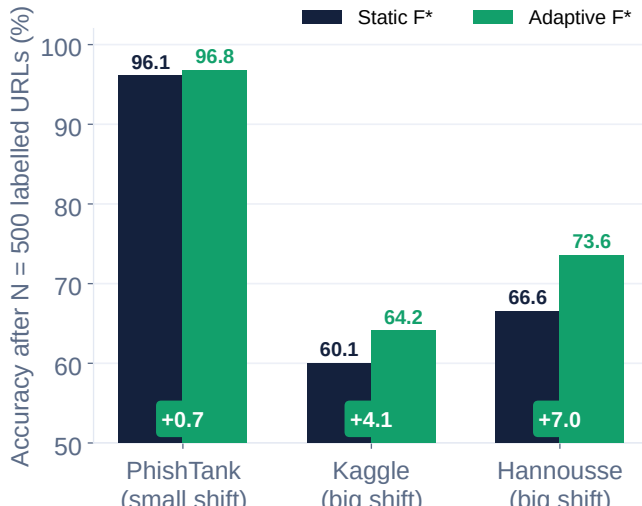
Old vs new phishing feature distributions — complements paper Fig. 4.

Appendix B — Compact stream summary bars



S1 (PhishTank) and S2 (cross-corpus): adaptive stays competitive / best.

Appendix C — Ablation accuracy curves



Adaptive UCB1 … r hard shift.